Customer Churn Prediction — Short Report

Supervised Machine Learning | Classification | IBM Telco Customer Churn (7,043 × 21)

1. Business Problem

A telecom company loses customers every month. Each lost customer means lost revenue and higher cost to acquire a replacement. The company needs a model that predicts which customers are likely to leave (**churn**) so the retention team can act early.

Target: Churn (Yes / No) — binary classification.
Main risk: False Negatives (predicting "stay" when the customer will leave) are costly. Therefore **Recall** and **F1** for the Churn class matter more than raw Accuracy.

- Contact high-risk customers with a personalized offer before they leave.
- Focus retention budget on customers the model ranks as most likely to churn.

2. Data and Findings

Item	Result
Rows / columns	7,043 × 21
Missing values	Few blanks in TotalCharges (converted to numeric)
Duplicates	None
Class balance	~73% stay (No), ~27% leave (Yes) — imbalanced

Main patterns (association, not causation)

1. **Month-to-month contracts** have a much higher churn rate than one-year or two-year contracts.
2. Customers who leave tend to have **shorter tenure** and **higher monthly charges**.

These patterns describe the data; they do not prove that contract type or price alone causes churn.

3. Method

Dropped identifier **customerID**. Train/test split 80/20 with **stratify** and fixed **random_state**. Preprocessing inside a **Pipeline** (median impute + scale for numeric; most-frequent impute + one-hot for categorical) — fit on training data only (no leakage). Compared **Baseline** (always predict majority class) + three models: Logistic Regression, Decision Tree, Random Forest. Tuned **Gradient Boosting** with GridSearchCV (F1 scoring, 5-fold CV on training data only). Final evaluation once on the held-out test set.

4. Results (test set)

Model	Accuracy	Precision	Recall	F1
Baseline (most_frequent)	~0.73	0.00	0.00	0.00
Logistic Regression	~0.81	~0.66	~0.56	~0.60
Decision Tree	~0.73	~0.49	~0.51	~0.50
Random Forest	~0.78	~0.60	~0.48	~0.53
Tuned Gradient Boosting	**~0.80**	**~0.66**	**~0.52**	**~0.58**

- Baseline looks accurate but catches **zero** churners.
- Tree models overfit (near-perfect train scores, weaker test scores).
- Logistic Regression and tuned Gradient Boosting generalise better.
- Confusion matrix: **TP** = correctly flagged churners; **FN** = missed churners (costliest); **FP** = false alarms; **TN** =

correctly left alone.

5. Business Recommendations

Actions for at-risk customers

1. Offer a discount, loyalty plan, or free add-on to customers predicted as high risk.
2. Encourage a switch from month-to-month to a longer contract with a small incentive.

Limitations

1. Class imbalance — the model can still miss some real churners.
2. Data is a snapshot; behaviour changes over time → retrain periodically.
3. Correlation is not causation; high charges may reflect richer packages.

Recommendation: Use the **tuned Gradient Boosting** model for operational scoring (best balance after proper validation). If the team needs a simple, explainable model for managers, use **Logistic Regression**.

End of report